NAME

mbmesh − reconstruct triangle meshes from swath bathymetry data

SYNOPSIS

mbmesh -I *datalist* [**-O** *outputdir*] [**-R** *west/east/south/north*] [**-L** *meters*] [**--metadata**] [*output options*] [**-V**] [**-H**]

DESCRIPTION

mbmesh reads swath sonar data through the MB-System datalist processing framework and reconstructs a triangle mesh from accepted bathymetry soundings. The program estimates local point spacing, derives default reconstruction settings, estimates oriented normals, reconstructs a signed scalar field with screened Poisson, extracts a surface with marching cubes, trims unsupported mesh regions, and writes the result as a binary glTF mesh.

The output mesh vertices are positioned in the same local Cartesian coordinate frame used for reconstruction, with units in meters. This is an east-north-up frame initialized from the first accepted ping's sensor position: X is positive east, Y is positive north, and Z is positive up relative to that origin. Longitude and latitude are converted to local meters using an equirectangular approximation based on the origin latitude.

This local frame is not the Earth-Centered, Earth-Fixed (ECEF) frame used by **mbgrd2gltf** by default, and it is not the ECEF-offset frame produced by **mbgrd2gltf --geoorigin**. The **--geoorigin** option in **mbgrd2gltf** allows vertices to be expressed as offsets from a local origin point rather than from Earth's center in order to reduce coordinate magnitudes and improve rendering precision. **mbmesh** currently uses its own local projected frame instead. When combining **mbmesh** GLB output with **mbgrd2gltf** output, transform both products into the same coordinate frame first; otherwise their vertices can appear offset or fail to line up.

The default output is:

mesh.glb
: Final triangle mesh in binary glTF format using the **mbmesh** local Cartesian coordinate frame.

Point cloud, diagnostic, and HTML viewer files are written only when requested with command-line output options.

OPTIONS

−I *datalist*, **--input** *datalist*
: Input MB-System datalist file. This option is required unless the datalist path is supplied as the first non-option argument.

−O *outputdir*, **--output** *outputdir*
: Output directory for generated files. The directory is created if needed. The default is *mbmesh_output*.

−R *west/east/south/north*, **--bounds** *west/east/south/north*
: Restrict accepted soundings to a geographic bounding box in degrees longitude and latitude.

−L *meters*, **--lod** *meters*, **--level-of-detail** *meters*
: Requested smallest feature size in local meters. If omitted, **mbmesh** derives level of detail from estimated point spacing. The implementation clamps the resolved value to a supported range and uses it to derive default normal-estimation, Poisson, and trimming parameters.

−-decimate *meters*
: Enable voxel-grid point decimation with the specified cell size in local meters. The legacy single-dash form **-decimate** is also accepted.

−-metadata, −-info
: Print dataset metadata and exit before mesh reconstruction. The metadata includes input counts, geodetic bounds, and estimated point spacing.

−V, −-verbose
Enable progress and diagnostic logging.

−H, −h, −-help
Display help message and exit.

OUTPUT OPTIONS

Every long option in this section also accepts a legacy single-dash form (e.g. **-html**, **-diagnostics**, **-all-outputs**) for backward compatibility.

−-html
Also write **mesh.html**, an X3DOM viewer page that references **mesh.glb**, launch a local Python HTTP server, and open the viewer in the default web browser.

−-xyz Also write both local and ECEF XYZ point cloud files.

−-local-xyz
Also write **pointcloud-local.xyz**, containing accepted and decimated soundings in the local Cartesian coordinate frame used for reconstruction. This is the same coordinate frame used by **mesh.glb**.

−-ecef-xyz
Also write **pointcloud-ecef.xyz**, containing accepted and decimated soundings in WGS84 Earth-Centered, Earth-Fixed coordinates.

−-oriented-ply
Also write **oriented-pointcloud-ecef.ply**, an oriented point cloud suitable for inspection.

−-pointcloud-glb
Also write **pointcloud.glb**, a diagnostic GLB point cloud for smaller datasets.

−-normal-glb
Also write **normals.glb**, diagnostic normal vectors for smaller datasets.

−-origin-glb
Also write **origins.glb**, diagnostic sensor-origin rays for smaller datasets.

−-raw-mesh-glb
Also write **raw_mesh.glb**, the untrimmed triangle mesh produced by marching cubes before support trimming.

−-diagnostics
Also write **pointcloud.glb**, **normals.glb**, and **origins.glb**.

−-all-outputs
Write all optional output files.

EXAMPLES

Reconstruct a mesh and write the default **mesh.glb**:
mbmesh -Idatalist.mb-1

Write the mesh, X3DOM viewer, and local/ECEF XYZ point clouds:
mbmesh -Idatalist.mb-1 -O output/mbmesh_run --html --xyz -V

Restrict processing to a geographic region and request a coarser level of detail:
mbmesh -Idatalist.mb-1 -R-122.5/-121.8/36.5/37.2 -L 0.5 -V

Print dataset metadata without reconstructing a mesh:
mbmesh -Idatalist.mb-1 --metadata

OUTPUT FILES

mbmesh always writes **mesh.glb**. Optional files are written only when their corresponding output options are supplied.

mesh.glb
Final trimmed triangle mesh in binary glTF format. Vertex positions are in the **mbmesh** local Cartesian coordinate frame.

raw_mesh.glb
Optional untrimmed triangle mesh before support trimming.

mesh.html
Optional X3DOM preview page referencing **mesh.glb**. When **--html** is used, **mbmesh** attempts to serve this page from the output directory and open it in the default web browser.

pointcloud-local.xyz
Optional XYZ point cloud in the same local Cartesian coordinate frame as **mesh.glb**.

pointcloud-ecef.xyz
Optional WGS84 Earth-Centered, Earth-Fixed XYZ point cloud.

oriented-pointcloud-ecef.ply
Optional oriented ECEF point cloud.

pointcloud.glb
Optional diagnostic point cloud GLB.

normals.glb
Optional diagnostic normal-vector GLB.

origins.glb
Optional diagnostic sensor-origin-ray GLB.

PROCESSING NOTES

Sounding filtering
mbmesh uses MB-System datalist processing and accepts valid bathymetry soundings. Geographic filtering is applied when **-R** is specified.

Coordinate frame
mbmesh initializes its local Cartesian frame from the first accepted ping's sensor position. Sounding elevation is stored as negative bathymetry, and sensor elevation is stored as negative sensor depth. The local-to-geodetic conversion used for optional ECEF point-cloud outputs reverses this equirectangular approximation before converting to WGS84 ECEF coordinates.

Automatic settings
The program estimates point spacing from accepted soundings and derives default normal-estimation, screened-Poisson, marching-cubes, and support trimming settings from the data and requested level of detail.

Support trimming
Unsupported mesh regions are removed after marching cubes. If trimming removes the entire mesh, **mbmesh** exits with an error rather than writing an empty surface.

MB-SYSTEM INTEGRATION

mbmesh integrates with the MB-System datalist processing framework and can read swath files in MB-System supported formats.

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Initial mbmesh implementation: CSUMB Capstone - Spring 2026

SEE ALSO

mb-system(1), **mbgrid**(1), **mbgrd2gltf**(1), **mbgrdviz**(1)